Exp No: 25 PRODUCER–CONSUMER PROBLEM USING SYSTEMC

BProgram:

```
// Code your testbench here.
// Uncomment the next line for SystemC modules.
// #include <systemc>
#include <systemc.h>
// Producer Module
SC_MODULE(Producer) {
sc_fifo_out<int> out;
void produce() {
for (int i = 1; i <= 5; i++) {
cout << "Produced: " << i << endl;
out.write(i);
wait(1, SC_SEC);
}
}
SC_CTOR(Producer) {
SC_THREAD(produce);
}
};
// Consumer Module
SC_MODULE(Consumer) {
sc_fifo_in<int> in;
void consume() {
int value;
while (true) {
in.read(value);
cout << "Consumed: " << value << endl;
wait(1, SC_SEC);
```

```
}
}
SC_CTOR(Consumer) {
SC_THREAD(consume);
}
};
int sc_main(int argc, char* argv[]) {
sc_fifo<int> fifo(5);
Producer producer("Producer");
Consumer consumer("Consumer");
producer.out(fifo);
consumer.in(fifo);
sc_start(10, SC_SEC);
return 0;
}
```

playground.com/x/Ezh5

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testbench.cpp design.

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wait(1, SC_SEC);
}
}
SC_CTOR(Producer) {
SC_THREAD(produce);
}
};
// Consumer Module
SC_MODULE(Consumer) {
sc_fifo_in<int> in;
void consume() {
int value;
while (true) {
```

Log Share

Log Share

```
        SystemC 2.3.3-Accellera --- May 23 2020 14:33:56
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        ALL RIGHTS RESERVED
Produced: 1
Consumed: 1
Produced: 2
Consumed: 2
Produced: 3
Consumed: 3
Produced: 4
Consumed: 4
Produced: 5
Consumed: 5
```